
Wine Assignment

Background Story:

You are working with a wine research institute in Italy that wants to automate the classification of wines into one of three cultivars based on their chemical composition. The wines come from the same region but were grown under different conditions and techniques, which result in subtle differences in chemical properties.

The institute has provided you with a dataset containing the results of a chemical analysis of wines grown in the same region. Your goal is to build and evaluate machine learning models that can accurately classify the wine cultivar.

Dataset: Wine Dataset (`sklearn.datasets.load_wine`)

- 178 samples
- 13 features
- Target: 3 wine cultivars (Class 0, 1, 2)

Instructions:

- Perform **Exploratory Data Analysis (EDA)** to understand the structure and distribution of the data.
- Train and evaluate **five classification models**:
  - K-Nearest Neighbors (KNN)
  - Logistic Regression
  - Decision Tree
  - Random Forest

- Support Vector Machine (You can apply different kernels)

- Use **Cross-Validation (CV)** where necessary .
- Use **Recursive Feature Elimination (RFE)** to select the most relevant features.

- Apply **Principal Component Analysis (PCA)** to reduce dimensionality where appropriate.

- Compare model results and present findings visually.

Objectives:

- As part of a wine classification audit, the quality control team wants to understand which chemical properties most strongly differentiate between the three wine cultivars.
- Lab testing of all chemical properties is expensive. Management wants to know if accurate classification is still possible using fewer chemical tests. Emulate a cost-saving initiative.
- The winery plans to automate wine type identification using machine learning. You are expected to recommend the best-performing model, give reasons
- The quality team needs a 2D visualization showing how well the wines naturally separate based on chemical properties. Provide visualization that can be used in internal reports to communicate the separation between wine cultivars.

Deliverables

1. **PowerPoint Presentation** (with 10+ visuals)

2. **Jupyter Notebook (**well documented code , with comments **)**

Submit here: https://forms.gle/CobS6CjJeA8pHA4k8